

AI-Powered Mobile Application for Cybercrime Reporting and Scam Detection in West Africa

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CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Cybercrime has become one of the defining security threats of our time. The expansion of digital connectivity around the world has opened the door to criminal activities that operate through electronic networks, and these activities have grown rapidly in both scale and complexity. According to the International Telecommunication Union (ITU), global cybercrime damages hit roughly \$8.4 trillion in 2022, and that number is expected to climb past \$10.5 trillion per year by 2025 (Morgan, 2022). Phishing, online fraud, identity theft, romance scams, financial exploitation; the list goes on. No continent, no sector, no demographic has been spared.

Africa, in particular, has seen a dramatic spike in cybercrime over the last decade. The continent is undergoing a rapid digital shift, driven largely by rising mobile phone use and widening internet access. This shift has created real opportunities for economic growth, but it has also left millions of new users exposed to cyber threats they may not fully understand. The African Union's Cybersecurity Report put the annual cost of cybercrime to African economies at around \$4 billion, a figure that represents a serious drag on socioeconomic progress (African Union, 2022). Within the continent, West Africa has emerged as a global hotspot for cybercriminal activity. Nigeria and Cameroon, in particular, rank among the hardest-hit nations (INTERPOL, 2021).

Nigeria sits at the center of Africa's cybercrime crisis. It is the continent's largest economy and most populous country, home to over 220 million people, and it has more than 100 million internet users, the biggest online population in Africa (Internet World Stats, 2023). The country has long been associated with advance-fee fraud, often called "419 scams" after the relevant section of the Nigerian Criminal Code. But the playbook has evolved. Today,

cybercriminals based in Nigeria are involved in business email compromise (BEC), romance scams, phishing campaigns, and cryptocurrency fraud. The Economic and Financial Crimes Commission (EFCC) reported that cybercrime-related losses in Nigeria topped \$500 million in 2022, with thousands of ordinary citizens victimized each year (EFCC, 2023). Yet even those numbers likely understate the problem. The Nigeria Computer Emergency Response Team (ngCERT) has pointed out that the vast majority of incidents are never reported, due to stigma, a lack of awareness about where to turn, and the absence of easy-to-use reporting platforms (ngCERT, 2022).

Cameroon tells a similar story, though with its own distinct characteristics. The country has roughly 15 million internet users, and its mobile money ecosystem is expanding fast. That growth has brought with it a surge in cybercrime, particularly mobile money fraud, SIM swap attacks, and social media scams (Ndifon & Achou, 2023). Cameroon's bilingual makeup, split between French-speaking and English-speaking populations, adds another layer of difficulty when it comes to public awareness campaigns. The National Agency for Information and Communication Technologies (ANTIC) has noted a steady increase in online fraud complaints, but Cameroon still lacks a centralized digital platform where citizens can report cybercrime or get immediate help (ANTIC, 2022). Making matters worse, the country's cybercrime legislation remains incomplete, leaving victims with few legal options.

To understand why cybercrime thrives in West Africa, you have to look at the broader context. Rapid urbanization, high rates of youth unemployment, and limited digital literacy have created conditions where cybercrime flourishes, both as a form of exploitation and, disturbingly, as something some young people see as a viable economic path (Warner, 2011). Many victims, especially those in lower-income communities, simply do not have the knowledge to spot scam

tactics, nor the financial cushion to recover from losses. Women, elderly individuals, and people going online for the first time are disproportionately targeted. Ironically, these same groups are the least likely to come forward and report what happened to them (Akanle et al., 2020).

Technology has a long track record of being applied to societal problems, and crime prevention is no exception. Mobile applications have been used across sectors to facilitate incident reporting, share information, and engage communities. In public safety specifically, mobile-based reporting systems have shown real promise in enabling faster responses and improving data collection for law enforcement (Bott & Young, 2012). But when it comes to cybercrime victim reporting in West Africa? That area remains remarkably underdeveloped.

Artificial Intelligence and Machine Learning offer genuine potential for tackling cybercrime detection and prevention. AI systems can sift through large volumes of data, pick out patterns tied to fraudulent activity, classify different scam types, and deliver real-time risk assessments (Buczak & Guven, 2016). In developed countries, machine learning has already proven its value in email filtering, financial fraud detection, and identifying phishing websites. The problem is that very little of this work has been adapted for mobile platforms built around the West African context, a context where mobile money fraud and locally tailored scam schemes are the dominant threats (Tade, 2022).

Cross-platform frameworks like Flutter now make it practical to build applications that work smoothly on both Android and iOS, which matters a great deal in markets where Android dominates. Pair that with real-time database solutions like Firebase, and developers have the tools to create responsive apps that handle dynamic data, secure authentication, and instant notifications (Google Developers, 2023). These capabilities provide the technical foundation needed to build an accessible cybercrime reporting tool for West African users.

There is also an ethical dimension worth considering. Combating cybercrime connects directly to one of the Millennium Project's 14 Global Challenges, specifically Challenge 12: How can transnational organized crime networks be stopped from becoming more powerful and sophisticated? Cybercrime is, by its nature, transnational. Perpetrators in West Africa routinely target victims in Europe, North America, and other parts of Africa, while also preying on local populations. Tackling this challenge calls for innovative, technology-driven approaches that give citizens the power to report incidents, access protective resources, and break the cycle of victimization (Glenn & Gordon, 2009).

This project sets out to develop an AI-powered mobile application that allows cybercrime victims in Nigeria and Cameroon to report scams, get instant risk assessments, and access educational resources. The app will use machine learning for scam pattern detection, Flutter for cross-platform development, and Firebase for real-time data management. The goal is to fill a critical gap in cybercrime victim support across West Africa and, in doing so, contribute to the broader effort of combating cybercrime on the continent.

1.2 Problem Statement

Even as cybercrime escalates across West Africa, victims face serious obstacles when they try to report scams or seek help. In Nigeria, the main channels for reporting are the EFCC and the Nigeria Police Force Cybercrime Unit. Both require in-person visits or formal written complaints, processes that are slow, intimidating, and effectively out of reach for many citizens, especially those living in rural areas (Adeniran, 2008). In Cameroon, ANTIC is the designated national cybersecurity body, but it has no public-facing digital reporting platform, and most citizens have never heard of it (ANTIC, 2022). The result is stark: INTERPOL's African Cyberthreat Assessment (2021) found that fewer than 20% of cybercrime incidents in Africa get

formally reported. In countries with limited digital reporting infrastructure, the actual figure is almost certainly lower.

Two existing solutions have tried to address parts of this problem. The first is ScamAdviser (www.scamadviser.com), an online tool that lets users check the trustworthiness of websites and online vendors before engaging with them. It serves a useful purpose for website verification, but it is primarily web-based, does not offer a dedicated mobile reporting function, and was never designed around the specific scam patterns common in West African markets, things like mobile money fraud, advance-fee schemes, SIM swap attacks, and phishing campaigns run through WhatsApp and SMS (ScamAdviser, 2023). The second is Trend Micro's Fraud Buster application, which uses AI to scan messages and URLs for potential threats. While effective at threat detection, it does not support victim reporting or resource provision, does not draw on region-specific scam data from West Africa, and lacks support for local languages such as Pidgin English, Yoruba, Igbo, or French, languages that are essential for reaching the people most at risk (Trend Micro, 2023).

Both solutions fall short in three important ways. First, neither provides an integrated platform that brings together scam reporting, AI-driven risk assessment, and victim resources in a single mobile application. Second, neither was built for the West African user, meaning they do not account for low-bandwidth connectivity, multilingual populations, mobile money fraud patterns, or the need to connect with local law enforcement. Third, neither uses machine learning models trained on West African cybercrime data, which is essential for detecting the region-specific scam patterns that make up the bulk of incidents in Nigeria and Cameroon.

The gap this project aims to fill is clear: there is no AI-powered, mobile-first platform that allows cybercrime victims in Nigeria and Cameroon to report incidents, receive intelligent

scam analysis, and access targeted support, all in one place. This underreporting crisis feeds directly into the growing power of transnational organized crime networks, as outlined in Global Challenge 12. When cybercriminals can operate with near-total impunity, and when law enforcement is starved of the data it needs to identify patterns, disrupt networks, and build cases, the problem only gets worse.

1.3 Project's Main Objective

This project aims to develop an AI-powered mobile application that allows cybercrime victims in Nigeria and Cameroon to report scams, receive instant risk assessments, and access educational and support resources. The ultimate goal is to improve cybercrime reporting rates and strengthen victim support infrastructure across West Africa.

1.3.1 Specific Objectives

1. To design and build a cross-platform mobile application using Flutter that gives cybercrime victims in Nigeria and Cameroon an intuitive, multilingual interface for reporting scam incidents, with a target development cycle of three months.

2. To train a machine learning model on West African cybercrime datasets that can classify reported scams into five categories, namely advance-fee fraud, mobile money fraud, phishing, romance scams, and identity theft, with at least 85% accuracy.

3. To integrate a real-time risk assessment feature, powered by Firebase, that provides users with instant feedback on the threat level of their reported incident within 30 seconds of submission.

4. To build an educational resource module within the app containing at least 20 articles, guides, and videos on scam identification and prevention, available in English, French, and Pidgin English.

5. To evaluate the app's usability and effectiveness through pilot testing with at least 50 users in Lagos, Nigeria, and Douala, Cameroon, collecting feedback to refine the system over a two-month testing period.

1.4 Research Questions

1. How can a mobile application be designed to offer an accessible, user-friendly platform for cybercrime victims in Nigeria and Cameroon to report scam incidents?

2. How accurately can machine learning algorithms detect and classify scam patterns specific to the West African context, including advance-fee fraud and mobile money scams?

3. How effective is a real-time, AI-powered risk assessment feature in helping users grasp the severity and nature of reported cybercrime incidents?

4. Which educational resources and support mechanisms are most effective in helping cybercrime victims in West Africa recover from scams and avoid future victimization?

5. How do pilot users in Lagos, Nigeria, and Douala, Cameroon, perceive the usability and usefulness of the application?

1.5 Project Scope

Geographically, this project focuses on two pilot locations: Lagos, Nigeria, and Douala, Cameroon. Lagos was chosen because it is Nigeria's commercial capital and largest city, home to over 20 million residents, the highest concentration of internet users in the country, and the epicenter of both cybercrime perpetration and victimization in West Africa (NCC, 2023).

Douala, Cameroon's economic hub with about 4 million residents, was selected for its rapidly growing mobile money ecosystem, its rising rates of digital fraud, and its value in representing the Francophone West African context (ANTIC, 2022). Testing in both cities allows the study to capture insights from Anglophone and Francophone populations alike.

The pilot testing population will consist of 50 to 100 adults (aged 18 and older) across both cities. These participants will be active mobile phone users who have either experienced cybercrime or face elevated risk of victimization. In Lagos, recruitment will take place in the Ikeja and Victoria Island areas, drawing from community technology hubs, university campuses, specifically the University of Lagos and Yaba College of Technology, and local NGOs focused on digital literacy. In Douala, participants will be recruited from the Akwa and Bonaberi neighborhoods through the University of Douala, cybercafé networks, and community organizations.

On the technical side, the application will be built with Flutter for cross-platform compatibility across Android and iOS, Firebase for real-time database management and user authentication, and Python-based machine learning libraries (TensorFlow and Scikit-learn) for the scam detection model. The model is scoped to five primary scam categories: advance-fee fraud (419 scams), mobile money fraud, phishing, romance scams, and identity theft. Training data will come from publicly available cybercrime datasets, supplemented with records from Nigerian and Cameroonian cybercrime reports, EFCC case files, and ANTIC incident logs.

The project is planned to run over six months: three months for application development and machine learning model training, followed by three months of pilot testing, evaluation, and iteration. The app will support English, French, and Pidgin English to ensure accessibility for both target populations. At this stage, the project does not include direct integration with law

enforcement databases, but the architecture is modular, designed so that future iterations could connect to EFCC, ANTIC, or INTERPOL systems.

1.6 Significance and Justification

If this application works as intended, it will give citizens in Nigeria and Cameroon a practical, technology-driven tool for reporting scam incidents and receiving immediate, intelligent feedback. That matters because the underreporting problem is severe. INTERPOL (2021) estimated that fewer than 20% of cybercrime cases in Africa are formally reported. A mobile-first, multilingual approach that removes the friction from reporting could meaningfully increase both the volume and quality of cybercrime data available to agencies like the EFCC and ANTIC, data they need to investigate and prosecute offenders more effectively.

The project also speaks to the Millennium Project's Global Challenge 12 on transnational organized crime. West African cybercriminal networks are among the most sophisticated on the continent, running operations that span multiple countries and target victims worldwide (Tade, 2022). A platform that detects scam patterns and aggregates structured incident data could serve as a foundational intelligence tool, useful for mapping cybercrime trends, identifying criminal networks, and supporting coordinated law enforcement action at national, regional, and continental levels (African Union, 2022).

Beyond enforcement, the app's educational module will equip vulnerable populations, including women, elderly users, and first-time internet adopters, with the knowledge to recognize and avoid scams, building a more digitally resilient society over time. The project stands to benefit multiple stakeholders: victims gain an accessible reporting channel and recovery resources; law enforcement agencies gain structured, analyzable incident data; researchers gain

access to cybercrime trend datasets; and policymakers gain the evidence they need to push for stronger cybersecurity legislation in both Nigeria and Cameroon.

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